

Technical Appendix: Dependence-Aware Evaluation of Panel Trading Strategy Evidence

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1 Purpose

This appendix contains proof details and secondary empirical artifacts that are omitted from the journal-facing main manuscript for length. The main manuscript is the claim boundary; this appendix is the audit trail for derivations, positive controls, robustness tables, and generated public-data artifacts.

2 Mean-Return Design-Effect Proof

For a selected row model $y_{t,i} = \mu + \lambda_t + \epsilon_{t,i}$ with constant selected count $\bar{n}$, same-date intraclass correlation ρ , and independent dates, the row-pooled mean equals the average of date means:

$$\hat{\mu} = \frac{1}{T\bar{n}} \sum_{t=1}^T \sum_{i=1}^{\bar{n}} y_{t,i} = \frac{1}{T} \sum_{t=1}^T \bar{y}_t.$$

The row-naive variance report is $\sigma^2/(T\bar{n})$. The true one-date variance is

$$\text{Var}(\bar{y}_t) = \frac{1}{\bar{n}^2} \{ \bar{n}\sigma^2 + \bar{n}(\bar{n}-1)\rho\sigma^2 \} = \frac{\sigma^2}{\bar{n}} \{ 1 + (\bar{n}-1)\rho \}.$$

Thus

$$\frac{se_{date,true}}{se_{row,naive}} = \sqrt{1 + (\bar{n}-1)\rho}.$$

This is the Moulton-like design effect used as the mean-return special case in the main text.

3 Sharpe UVIF Derivation

Let R_t be the date-level portfolio excess return and $G_t = (R_t, R_t^2)'$. The annualized Sharpe ratio is the smooth functional

$$f(\mu, m_2) = \sqrt{252} \frac{\mu}{(m_2 - \mu^2)^{1/2}}, \quad d_R = \nabla f(\mu_R, m_{2,R}).$$

On sets where $v_R = m_{2,R} - \mu_R^2 > 0$, the first-order expansion is

$$\widehat{\text{SR}}_R - \text{SR}_R = d'_R(\hat{\theta}_R - \theta_R) + o_p(T_R^{-1/2}), \quad \theta_R = \mathbb{E}[G_t].$$

With long-run covariance $\Omega_R = \Gamma_0^R + \sum_{k \geq 1} (\Gamma_k^R + \Gamma_k^{R'})$,

$$\text{Var}_{LR}(\widehat{\text{SR}}_R) = T_R^{-1} d'_R \Omega_R d_R + o(T_R^{-1}).$$

The row-pooled IID calculation applies the same delta method to $\bar{n}T_R$ row observations $H_{t,i} = (Y_{t,i}, Y_{t,i}^2)'$:

$$\text{Var}_{IID, \text{row}}(\widehat{\text{SR}}_{\text{row}}) = (\bar{n}T_R)^{-1} d'_Y \Psi_Y d_Y + o((\bar{n}T_R)^{-1}).$$

Taking the ratio gives the Sharpe UVIF:

$$\text{UVIF}_{\text{SR}} = \frac{\bar{n} d'_R \Omega_R d_R}{d'_Y \Psi_Y d_Y} = \frac{\bar{n} d'_R \Gamma_0^R d_R}{d'_Y \Psi_Y d_Y} \times \frac{d'_R \Omega_R d_R}{d'_R \Gamma_0^R d_R}.$$

The first term is the cross-sectional component and the second term is the serial long-run factor. Componentwise,

$$d'_R \Omega_R d_R = \frac{252}{v_R^3} \left[m_{2,R}^2 \Omega_{11} - \frac{\mu_R m_{2,R}}{2} (\Omega_{12} + \Omega_{21}) + \frac{\mu_R^2}{4} \Omega_{22} \right].$$

The Ω_{22} term contains long-run covariances of squared returns and therefore brings fourth moments into Sharpe-ratio variance.

4 Capacity Frontier Proof

For a square-root impact diagnostic, the dependence-adjusted one-sided lower confidence margin for gross mean return is

$$\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T},$$

where σ_{row} is the standard deviation of row-pooled returns before date aggregation or UVIF adjustment. Square-root impact subtracts $\lambda \widehat{TO} \sqrt{A/V}$. Requiring the net lower

Table 1: Synthetic positive-control audit.

Panel	$\bar{\rho}$	Adj.	SR	HAC p^+	RW p	Perm. p^+
Synthetic control	0.202	4.58	1.744	7.67e-13	5.00e-04	9.99e-04

margin to be nonnegative gives

$$\widehat{\lambda TO} \sqrt{A/V} \leq \bar{r}_{\text{gross}} - z_{\alpha} \mathcal{M} \sigma_{\text{row}} / \sqrt{T}.$$

Rearranging yields the capacity frontier reported in the main text. If the volatility input is already the date-level portfolio volatility, the date boundary has already embedded the cross-sectional UVIF component, so the cross-sectional part of $\mathcal{M}$ must not be multiplied in again.

5 Higher-Moment UVIF and EIR Details

For $\theta^{(4)} = (\mu, m_2, m_3, m_4)'$, define skewness and kurtosis from the central moments implied by $\theta^{(4)}$. Any adjusted Sharpe functional $h(\theta^{(4)})$ has a delta-method variance $\nabla h' \Omega^{(4)} \nabla h$, where $\Omega^{(4)}$ is the long-run covariance matrix of $(R_t, R_t^2, R_t^3, R_t^4)'$. The higher-moment UVIF is the corresponding ratio of date-level long-run variance to row-pooled IID variance. It is a diagnostic extension, not an additional empirical rejection gate.

The EIR diagnostic is

$$\text{EIR}_t = \frac{I(S_t; R_t)}{n_t \bar{I}_t}, \quad \bar{I}_t = n_t^{-1} \sum_{i=1}^{n_t} I(s_{t,i}; r_{t+1,i}).$$

Under the same equicorrelated Gaussian approximation used for the mean-return design effect,

$$\text{EIR}_t(\rho) \approx \frac{1}{1 + (n_t - 1)\rho}.$$

Thus an EIR of 0.111 means that roughly $1 - 0.111 = 0.889$, or 89 percent, of nominal row-level information is redundant under the diagnostic approximation.

Table 2: Pre-registered public candidate evaluation campaign.

Candidate	status	SR	HAC p+	Boot p+	RW p	Perm p	Net SR	Pass 5%
french_momentum_deciles_daily_rank	ok	0.447	7.54e-05	2.00e-04	2.00e-04	1.000	0.447	no
french_momentum_deciles_daily_wml	ok	0.497	5.53e-06	2.00e-04	2.00e-04	1.00e-04	0.497	yes
french_size_bm_daily_dynamic_momentum	ok	0.036	0.445	0.458	0.516	0.947	-0.072	no
stooq_fixed_etf_daily_dynamic_momentum	failed							no
aqr_bab_equity_factors_daily	ok	0.741	3.08e-13	2.00e-04	2.00e-04		0.741	no
aqr_qmj_factors_daily	ok	0.536	2.26e-05	2.00e-04	2.00e-04		0.536	no
aqr_hml_devil_factors_daily	ok	0.336	0.004	0.002	0.002		0.336	no
aqr_value_momentum_everywhere_monthly	ok	2.623	1.62e-04	2.00e-04	2.00e-04		2.623	no

Table 3: Canonical French momentum benchmark validation.

Candidate	thr	n	start	end	panel SR	direct WML SR	scale
french_momentum_deciles_daily_wml	0.900	26131	1926-11-04	2026-04-30	0.497	0.497	0.500

6 Synthetic Positive Control

7 Full Empirical Artifacts

8 Full Monte Carlo Artifacts

9 Robustness and Registry Notes

The public registry is frozen in `code/campaign_registry.json`. It records loaders, public source metadata, threshold menus, primary thresholds, and lookback/skip parameters. Changing a source, lookback, threshold, cost assumption, or eligibility rule after seeing results creates a new researcher menu and must be reported as exploratory.

Stationarity diagnostics, holdout splits, HAC bandwidth sweeps, prewhitened HAC, fixed-fraction HAC, moving-block bootstrap, and stationary bootstrap results are secondary robustness checks. They are used to identify fragile claims, not to replace the main date-return decision boundary.

Table 4: Sampling-boundary design-effect diagnostics.

Candidate	status	thr	rows	dates	avg names	rho	SE infl.	row t	HAC z	row p+	HAC p+	RW p
aqr_bab_equity_factors_daily	not_applicable_single_series	0.000	25281	25281	1.000							2.00e-04
aqr_hml_devil_factors_daily	not_applicable_single_series	0.000	26590	26590	1.000							0.002
aqr_qmj_factors_daily	not_applicable_single_series	0.000	17681	17681	1.000							2.00e-04
aqr_value_momentum_everywhere_monthly	not_applicable_single_series	0.000	651	651	1.000							2.00e-04
french_momentum_deciles_daily_rank	ok	0.700	104524	26131	4.000	-0.138	0.766	3.584	3.790	1.70e-04	7.54e-05	2.00e-04
french_momentum_deciles_daily_wml	ok	0.900	52262	26131	2.000	-0.583	0.646	3.299	4.395	4.85e-04	5.53e-06	2.00e-04
french_size_bm_daily_dynamic_momentum	ok	0.500	45201	3477	13.000	0.667	3.000	0.664	0.222	0.253	0.412	0.516

Table 5: Artifact-backed phantom-alpha audit summary for panel candidates.

Candidate	Row p+	UVIF	EIR	Audit p+	Net SR	Status
french_momentum_deciles_daily_rank	1.70e-04	1.000	1.000	1.000	0.447	phantom
french_momentum_deciles_daily_wml	4.85e-04	1.000	1.000	2.00e-04	0.497	robust
french_size_bm_daily_dynamic_momentum	0.253	9.002	0.111	0.947	-0.072	cost fail

Table 6: Horizon-effect audit for the dynamic Size/BM momentum panel.

L	thr	dates	avg names	gross SR	net SR 5bps	turnover	UVIF SR	serial UVIF	HAC p+
1	0.500	3497	12.913	0.073	-0.394	0.733	11.498	0.891	0.386
5	0.500	3493	13.000	-0.312	-0.528	0.339	12.880	0.990	0.878
21	0.500	3477	13.000	0.058	-0.050	0.169	12.121	0.933	0.412

Table 7: Primary-threshold dependence-aware Sharpe inference.

Candidate	thr	Method	SR	SE	CI low	CI high	p+	n
aqr_bab_equity_factors_daily	0.000	block	0.741	0.106	0.533	0.951	2.00e-04	25281
aqr_bab_equity_factors_daily	0.000	HAC	0.741	0.103	0.540	0.943	3.08e-13	25281
aqr_bab_equity_factors_daily	0.000	prewhite	0.741	0.104	0.538	0.945	4.38e-13	25281
aqr_hml_devil_factors_daily	0.000	block	0.336	0.114	0.105	0.552	0.002	26590
aqr_hml_devil_factors_daily	0.000	HAC	0.336	0.126	0.089	0.583	0.004	26590
aqr_hml_devil_factors_daily	0.000	prewhite	0.336	0.111	0.119	0.553	0.001	26590
aqr_qmj_factors_daily	0.000	block	0.536	0.133	0.276	0.795	2.00e-04	17681
aqr_qmj_factors_daily	0.000	HAC	0.536	0.131	0.279	0.794	2.26e-05	17681
aqr_qmj_factors_daily	0.000	prewhite	0.536	0.128	0.285	0.787	1.40e-05	17681
aqr_value_momentum_everywhere_monthly	0.000	block	2.623	0.716	1.230	4.041	2.00e-04	651
aqr_value_momentum_everywhere_monthly	0.000	HAC	2.625	0.730	1.194	4.056	1.62e-04	651
aqr_value_momentum_everywhere_monthly	0.000	prewhite	2.625	0.723	1.207	4.043	1.42e-04	651
french_momentum_deciles_daily_rank	0.700	block	0.447	0.114	0.222	0.667	2.00e-04	26131
french_momentum_deciles_daily_rank	0.700	HAC	0.447	0.118	0.216	0.679	7.54e-05	26131
french_momentum_deciles_daily_rank	0.700	prewhite	0.447	0.114	0.224	0.670	4.18e-05	26131
french_momentum_deciles_daily_wml	0.900	block	0.497	0.111	0.275	0.708	2.00e-04	26131
french_momentum_deciles_daily_wml	0.900	HAC	0.497	0.113	0.275	0.718	5.53e-06	26131
french_momentum_deciles_daily_wml	0.900	prewhite	0.497	0.111	0.279	0.715	3.99e-06	26131
french_size_bm_daily_dynamic_momentum	0.500	block	0.036	0.261	-0.487	0.530	0.458	3477
french_size_bm_daily_dynamic_momentum	0.500	HAC	0.036	0.260	-0.473	0.544	0.445	3477

Table 8: Same-date signal-permutation diagnostics for panel candidates.

Candidate	thr	obs SR	null mean	null sd	p+	perms
french_momentum_deciles_daily_rank	0.000	0.395	0.410	0.039	0.646	10000
french_momentum_deciles_daily_rank	0.300	0.419	0.434	0.024	0.733	10000
french_momentum_deciles_daily_rank	0.500	0.419	0.434	0.024	0.733	10000
french_momentum_deciles_daily_rank	0.700	0.447	0.447	1.11e-16	1.000	10000
french_momentum_deciles_daily_rank	0.900	0.497	0.404	0.042	0.015	10000
french_momentum_deciles_daily_wml	0.900	0.497	-0.001	0.098	1.00e-04	10000
french_size_bm_daily_dynamic_momentum	0.000	0.011	0.052	0.017	0.993	10000
french_size_bm_daily_dynamic_momentum	0.300	0.010	0.072	0.023	0.996	10000
french_size_bm_daily_dynamic_momentum	0.500	0.036	0.083	0.029	0.947	10000
french_size_bm_daily_dynamic_momentum	0.700	0.061	0.112	0.040	0.893	10000
french_size_bm_daily_dynamic_momentum	0.900	0.025	0.098	0.069	0.855	10000

Table 9: HAC bandwidth, prewhitening, and fixed-b robustness diagnostics.

Candidate	Diagnostic	SR	SE	p+
aqr_bab_equity_factors_daily	HAC K=auto	0.741	0.103	3.08e-13
aqr_bab_equity_factors_daily	HAC K=21	0.741	0.119	2.04e-10
aqr_bab_equity_factors_daily	HAC K=63	0.741	0.130	6.06e-09
aqr_bab_equity_factors_daily	prewhite	0.741	0.104	4.38e-13
aqr_bab_equity_factors_daily	fixed-b 0.1	0.741	0.166	0.002
aqr_hml_devil_factors_daily	HAC K=auto	0.336	0.126	0.004
aqr_hml_devil_factors_daily	HAC K=21	0.336	0.128	0.004
aqr_hml_devil_factors_daily	HAC K=63	0.336	0.137	0.007
aqr_hml_devil_factors_daily	prewhite	0.336	0.111	0.001
aqr_hml_devil_factors_daily	fixed-b 0.1	0.336	0.110	0.004
aqr_qmj_factors_daily	HAC K=auto	0.536	0.131	2.26e-05
aqr_qmj_factors_daily	HAC K=21	0.536	0.140	6.33e-05
aqr_qmj_factors_daily	HAC K=63	0.536	0.147	1.32e-04
aqr_qmj_factors_daily	prewhite	0.536	0.128	1.40e-05
aqr_qmj_factors_daily	fixed-b 0.1	0.536	0.151	0.002
aqr_value_momentum_everywhere_monthly	HAC K=auto	2.625	0.730	1.62e-04
aqr_value_momentum_everywhere_monthly	HAC K=21	2.625	0.712	1.14e-04
aqr_value_momentum_everywhere_monthly	HAC K=63	2.625	0.659	3.41e-05
aqr_value_momentum_everywhere_monthly	prewhite	2.625	0.723	1.42e-04
aqr_value_momentum_everywhere_monthly	fixed-b 0.1	2.625	0.656	0.004
french_momentum_deciles_daily_rank	HAC K=auto	0.447	0.118	7.54e-05
french_momentum_deciles_daily_rank	HAC K=21	0.447	0.120	9.94e-05
french_momentum_deciles_daily_rank	HAC K=63	0.447	0.122	1.21e-04
french_momentum_deciles_daily_rank	prewhite	0.447	0.114	4.18e-05
french_momentum_deciles_daily_rank	fixed-b 0.1	0.447	0.143	0.004
french_momentum_deciles_daily_wml	HAC K=auto	0.497	0.113	5.53e-06
french_momentum_deciles_daily_wml	HAC K=21	0.497	0.116	9.62e-06
french_momentum_deciles_daily_wml	HAC K=63	0.497	0.117	1.01e-05
french_momentum_deciles_daily_wml	prewhite	0.497	0.111	3.99e-06
french_momentum_deciles_daily_wml	fixed-b 0.1	0.497	0.147	0.002

Table 10: Composite gate sensitivity by alpha threshold.

alpha	candidates	passes
0.010	8	1
0.050	8	1

Table 11: Holdout and subperiod Sharpe diagnostics.

Candidate	window	SR	n
aqr_bab_equity_factors_daily	train_70pct	0.887	17696
aqr_bab_equity_factors_daily	holdout_30pct	0.531	7585
aqr_bab_equity_factors_daily	quarter_min	0.280	6320
aqr_bab_equity_factors_daily	quarter_max	1.270	6320
aqr_hml_devil_factors_daily	train_70pct	0.465	18613
aqr_hml_devil_factors_daily	holdout_30pct	0.120	7977
aqr_hml_devil_factors_daily	quarter_min	0.273	6647
aqr_hml_devil_factors_daily	quarter_max	0.445	6647
aqr_qmj_factors_daily	train_70pct	0.794	12376
aqr_qmj_factors_daily	holdout_30pct	0.305	5305
aqr_qmj_factors_daily	quarter_min	0.174	4421
aqr_qmj_factors_daily	quarter_max	1.083	4420
aqr_value_momentum_everywhere_monthly	train_70pct	2.867	455
aqr_value_momentum_everywhere_monthly	holdout_30pct	1.974	196
aqr_value_momentum_everywhere_monthly	quarter_min	1.707	163
aqr_value_momentum_everywhere_monthly	quarter_max	4.313	163
french_momentum_deciles_daily_rank	train_70pct	0.615	18292
french_momentum_deciles_daily_rank	holdout_30pct	0.229	7839
french_momentum_deciles_daily_rank	quarter_min	0.171	6532
french_momentum_deciles_daily_rank	quarter_max	1.280	6533
french_momentum_deciles_daily_wml	train_70pct	0.641	18292
french_momentum_deciles_daily_wml	holdout_30pct	0.310	7839
french_momentum_deciles_daily_wml	quarter_min	0.224	6533
french_momentum_deciles_daily_wml	quarter_max	1.370	6533
french_size_bm_daily_dynamic_momentum	train_70pct	0.050	2428
french_size_bm_daily_dynamic_momentum	holdout_30pct	0.005	1049
french_size_bm_daily_dynamic_momentum	quarter_min	-0.657	853
french_size_bm_daily_dynamic_momentum	quarter_max	0.211	875

Table 12: Turnover-scaled cost sensitivity and break-even cost.

Candidate	cost bps	turnover	gross SR	net SR	break-even bps
aqr_bab_equity_factors_daily	0	0.000	0.741	0.741	
aqr_bab_equity_factors_daily	1	0.000	0.741	0.741	
aqr_bab_equity_factors_daily	5	0.000	0.741	0.741	
aqr_bab_equity_factors_daily	10	0.000	0.741	0.741	
aqr_bab_equity_factors_daily	25	0.000	0.741	0.741	
aqr_hml_devil_factors_daily	0	0.000	0.336	0.336	
aqr_hml_devil_factors_daily	1	0.000	0.336	0.336	
aqr_hml_devil_factors_daily	5	0.000	0.336	0.336	
aqr_hml_devil_factors_daily	10	0.000	0.336	0.336	
aqr_hml_devil_factors_daily	25	0.000	0.336	0.336	
aqr_qmj_factors_daily	0	0.000	0.536	0.536	
aqr_qmj_factors_daily	1	0.000	0.536	0.536	
aqr_qmj_factors_daily	5	0.000	0.536	0.536	
aqr_qmj_factors_daily	10	0.000	0.536	0.536	
aqr_qmj_factors_daily	25	0.000	0.536	0.536	
aqr_value_momentum_everywhere_monthly	0	0.000	2.623	2.623	
aqr_value_momentum_everywhere_monthly	1	0.000	2.623	2.623	
aqr_value_momentum_everywhere_monthly	5	0.000	2.623	2.623	
aqr_value_momentum_everywhere_monthly	10	0.000	2.623	2.623	
aqr_value_momentum_everywhere_monthly	25	0.000	2.623	2.623	
french_momentum_deciles_daily_rank	0	1.91e-05	0.447	0.447	85893.000
french_momentum_deciles_daily_rank	1	1.91e-05	0.447	0.447	85893.000
french_momentum_deciles_daily_rank	5	1.91e-05	0.447	0.447	85893.000
french_momentum_deciles_daily_rank	10	1.91e-05	0.447	0.447	85893.000
french_momentum_deciles_daily_rank	25	1.91e-05	0.447	0.447	85893.000
french_momentum_deciles_daily_wml	0	1.91e-05	0.497	0.497	122232.000
french_momentum_deciles_daily_wml	1	1.91e-05	0.497	0.497	122232.000
french_momentum_deciles_daily_wml	5	1.91e-05	0.497	0.497	122232.000
french_momentum_deciles_daily_wml	10	1.91e-05	0.497	0.497	122232.000
french_momentum_deciles_daily_wml	25	1.91e-05	0.497	0.497	122232.000
french_size_bm_daily_dynamic_momentum	0	0.169	0.036	0.036	
french_size_bm_daily_dynamic_momentum	1	0.169	0.036	0.014	
french_size_bm_daily_dynamic_momentum	5	0.169	0.036	-0.072	
french_size_bm_daily_dynamic_momentum	10	0.169	0.036	-0.179	
french_size_bm_daily_dynamic_momentum	25	0.169	0.036	-0.502	

Table 13: Simulation DGP parameter grid.

DGP	Name	T	N	mu	sigma	AR1	rho	GARCH a	GARCH b	K	regime
01_iid	IID baseline	1000	50	4.00e-04	0.015	0.000	0.000	0.000	0.000	1	no
02_ser_mild	Serial mild	1000	50	4.00e-04	0.015	0.200	0.000	0.000	0.000	1	no
03_ser_strong	Serial strong	1000	50	4.00e-04	0.015	0.500	0.000	0.000	0.000	1	no
04_xs_mild	Cross mild	1000	50	4.00e-04	0.015	0.000	0.200	0.000	0.000	1	no
05_xs_strong	Cross strong	1000	50	4.00e-04	0.015	0.000	0.500	0.000	0.000	1	no
06_both_mod	Both moderate	1000	50	4.00e-04	0.015	0.200	0.200	0.000	0.000	1	no
07_both_strong	Both strong	1000	50	4.00e-04	0.015	0.500	0.500	0.000	0.000	1	no
08_realistic	Realistic	2500	100	4.00e-04	0.015	0.150	0.350	0.000	0.000	1	no
09_garch_mild	GARCH mild	1000	50	4.00e-04	0.015	0.200	0.200	0.050	0.900	1	no
10_garch_strong	GARCH strong	1000	50	4.00e-04	0.015	0.300	0.400	0.100	0.850	1	no
11_multifactor	Multi-factor K=3	1000	50	4.00e-04	0.015	0.200	0.300	0.000	0.000	3	no
12_regime	Regime-switching	2000	50	4.00e-04	0.015	0.150	0.250	0.000	0.000	1	yes
13_high_ar1	High-persistence AR(1)	1000	50	4.00e-04	0.015	0.700	0.300	0.000	0.000	1	no
14_garch_highvol	High-volatility GARCH	1000	50	4.00e-04	0.015	0.200	0.300	0.150	0.800	1	no

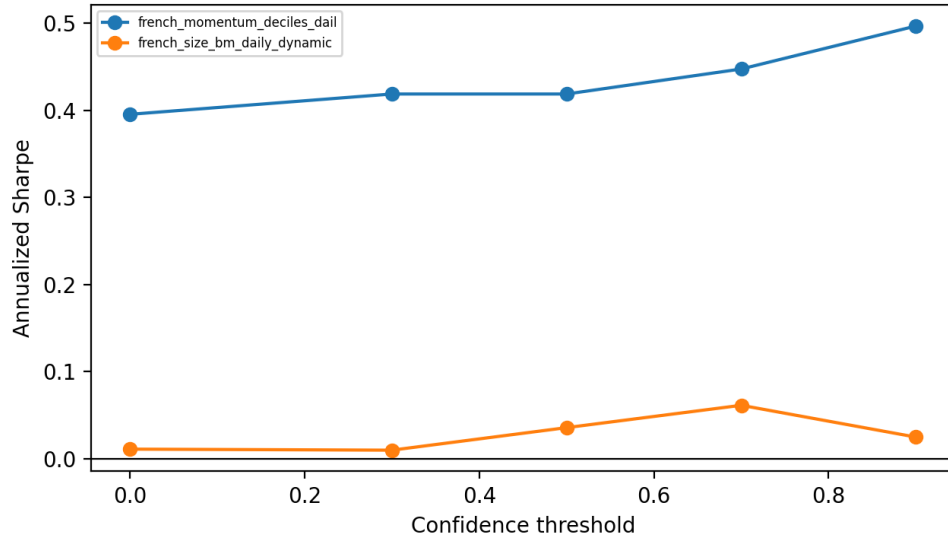


Figure 1: Threshold profile for public candidate panels.

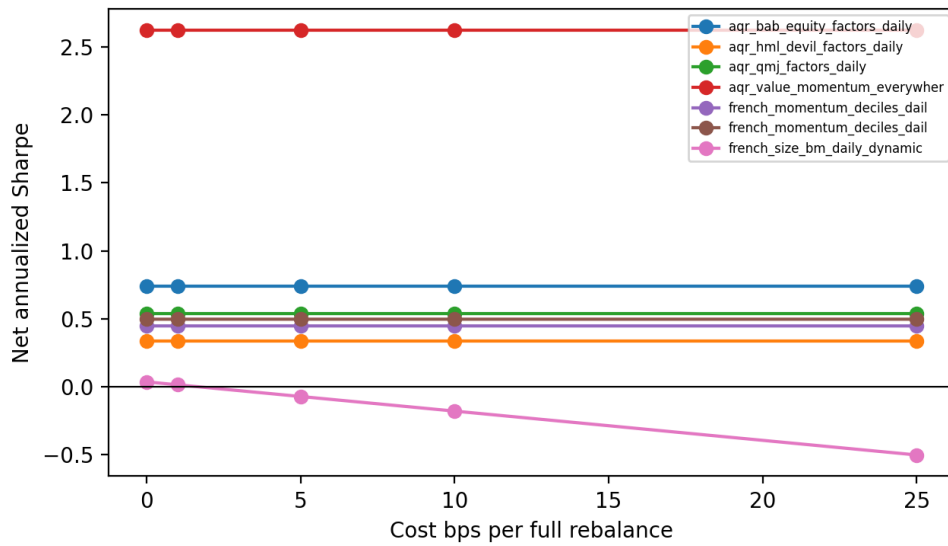


Figure 2: Turnover-scaled cost sensitivity at primary candidate thresholds.

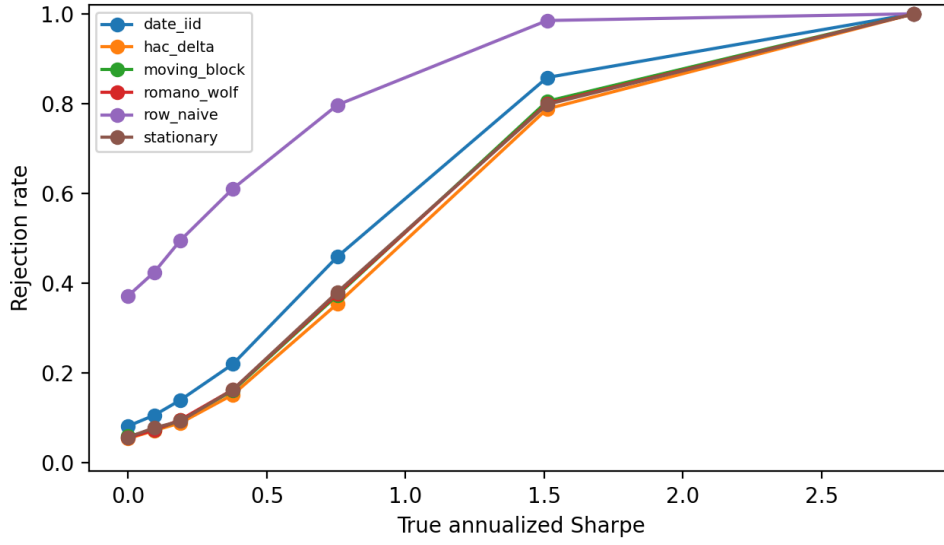


Figure 3: Monte Carlo rejection rates by true annualized Sharpe.

Table 14: Monte Carlo 95 percent interval coverage.

DGP	blocked	hac_delta	iid	row_naive	stationary
01_iid	0.947	0.952	0.951	0.000	0.947
02_ser_mild	0.938	0.943	0.898	0.000	0.936
03_ser_strong	0.921	0.934	0.734	0.000	0.919
04_xs_mild	0.950	0.950	0.952	0.073	0.950
05_xs_strong	0.934	0.938	0.942	0.262	0.938
06_both_mod	0.937	0.939	0.898	0.088	0.936
07_both_strong	0.934	0.943	0.771	0.178	0.935
08_realistic	0.937	0.941	0.901	0.107	0.937
09_garch_mild	0.926	0.940	0.877	0.110	0.930
10_garch_strong	0.926	0.948	0.862	0.199	0.927
11_multifactor	0.937	0.947	0.895	0.225	0.940
12_regime	0.929	0.934	0.890	0.306	0.929
13_high_ar1	0.899	0.933	0.562	0.137	0.907
14_garch_highvol	0.940	0.948	0.895	0.193	0.941

Table 15: Null rejection rates at nominal 5 percent size.

dgp	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
null_dep	0.094	0.066	0.070	0.068	0.352	0.067
null_garch	0.098	0.063	0.070	0.067	0.351	0.068
null_iid	0.060	0.060	0.061	0.060	0.062	0.062

Table 16: Power by true annualized Sharpe.

index	true_sharpe	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
0.000	0.000	0.081	0.053	0.058	0.055	0.371	0.056
1.000	0.094	0.105	0.071	0.072	0.072	0.424	0.077
2.000	0.189	0.139	0.088	0.094	0.095	0.495	0.092
3.000	0.378	0.219	0.151	0.160	0.162	0.610	0.162
4.000	0.755	0.459	0.354	0.373	0.376	0.797	0.380
5.000	1.511	0.858	0.788	0.805	0.800	0.985	0.798
6.000	2.833	1.000	1.000	1.000	1.000	1.000	1.000